Project 3 Report:

Introudction:

Understanding the three-dimensional structure of a protein is essential for explaining its function, interactions, and role in biological systems. Because protein activity is largely determined by its spatial conformation rather than its amino acid sequence alone, accurate structure prediction has been one of the central challenges in computational biology. For decades, experimental techniques such as X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy have served as the “gold standard” for structure determination. However, these methods are time-consuming, expensive, and not always feasible for every protein.

Recent advances in artificial intelligence have dramatically transformed the field of protein structure prediction. In particular, AlphaFold2 and its successor AlphaFold3, developed by DeepMind, have achieved unprecedented accuracy in predicting protein structures directly from amino acid sequences.These systems leverage deep learning architectures trained on large databases of known protein structures to infer geometric constraints, residue–residue interactions, and ultimately full three-dimensional conformations.

In this project, we predict the structures of two protein sequences using AlphaFold and compare them to their experimentally determined structures in the Protein Data Bank (PDB). Sequence 1 corresponds to a fimbrial adhesin from *Proteus mirabilis*, whose structure is available as 6Y4F chain A. Sequence 2 corresponds to a fragment of the human CST complex subunit CTC1, with its experimental structure deposited as 6W6W chain B.

The goal of this report is to evaluate how closely AlphaFold’s predicted models match the experimental structures and to analyze any similarities or differences in overall fold and structural features

Results & Analysis:

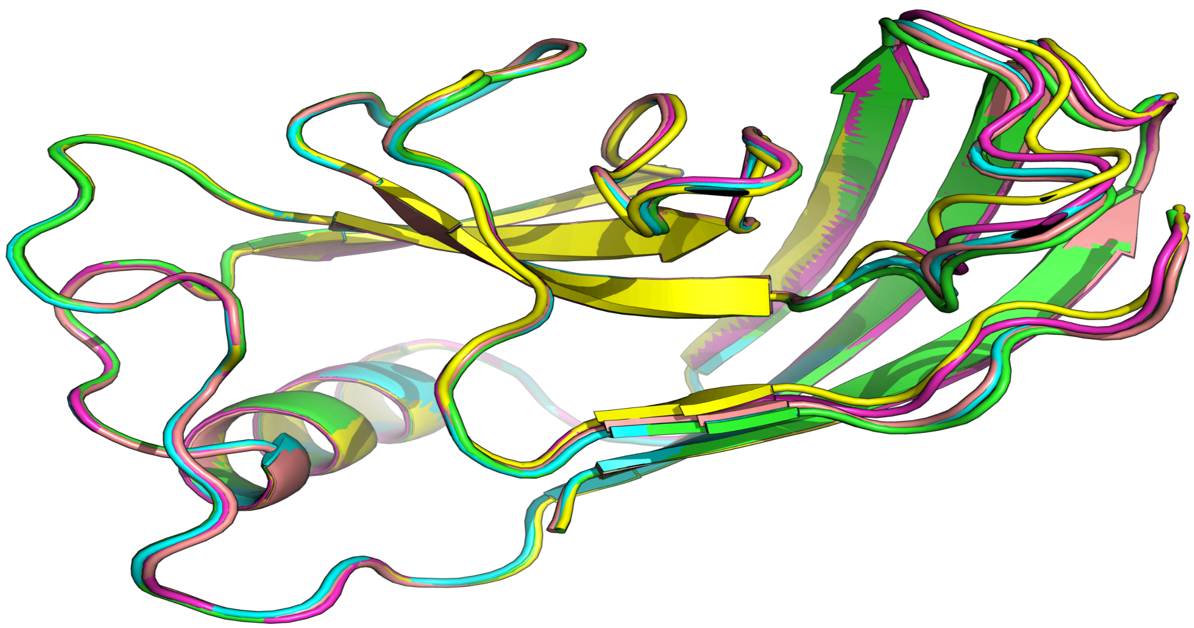
Note:The matrixes show the Root-Mean-Square Deviation (RMSD) values between five predicted structures (S1–S5) and the Gold Standard (GS), which represents the actual structure. RMSD measures the average distance between corresponding atoms after structural alignment; lower values indicate greater structural similarity. It's also given in terms of Angstrom Units(Å).

6yf4:

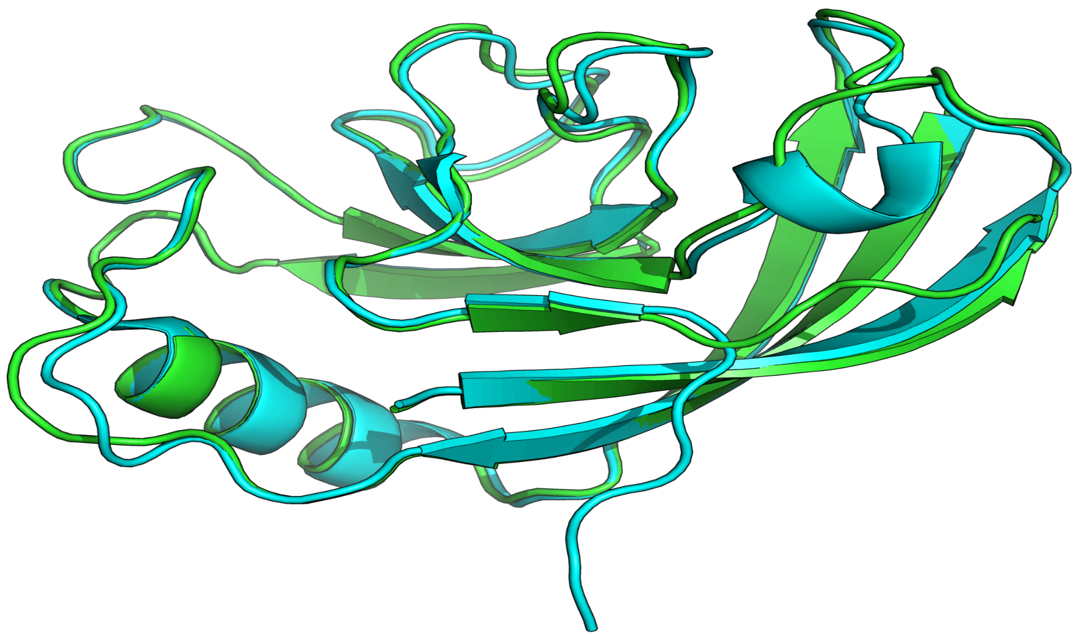
> Fimbrial adhesin|Proteus mirabilis (strain HI4320) (529507)
Sequence:SIFSYITESTGTPSNATYTYVIERWDPETSGILNPCYGWPVCYVTVNHKHTVNG TGGNPAFQIARIEKLRTLAEVRDVVLKNRSFPIEGQTTHRGPSLNSNQECVGLFYQPNSSG ISPRGK LLPGSLCGIAPPP

Matrix obtained from alignment of all 5 predicted sequences with each other and the given Gold Standard. Note All values are the Root-Mean-Square-Deviation(RMSD) measurement which uses Angstrom(Å) units:

	S1	S2	S3	S4	S5	GS
S1	0	**.207**	**.251**	**.265**	**.214**	**.511**
S2	.207	0	**.232**	**.268**	**.197**	**.568**
S3	.251	.232	0	**.195**	**.178**	**.672**
S4	.265	.268	.195	0	**.225**	**.653**
S5	.214	.197	.178	.225	0	**.546**



Caption: This image displays the superimposition of structures S1–S5 aligned on top of one another, clearly illustrating their nearly identical structural similarity. The near-perfect overlap among the models visually confirms the strong precision observed in the RMSD analysis, demonstrating that the predicted structures are highly consistent with each other.



Caption:This image shows the best match (S1) aligned with the Gold Standard structure. Despite its high precision, the poor overlap highlights its low accuracy and large structural deviation.

Precision:

The RMSD values among S1–S5 are extremely low, ranging from 0.178 to 0.268 Å. Because all values are well below 1 Å, this indicates that the five predicted models are nearly identical to one another. Such minimal structural deviation demonstrates very high precision, meaning the prediction method consistently generates almost the same structure across independent runs.For instance, the smallest deviation occurs between S3 and S5 (0.178 Å), reflecting near-perfect structural agreement. Even the largest deviation among predictions (S2 vs. S5 at 0.268 Å) remains very small, further confirming strong internal consistency. Overall, the tight clustering of RMSD values shows excellent reproducibility and stability in the modeling process.

Accuracy:

When compared to the Gold Standard structure, the predicted models show RMSD values ranging from 0.511 to 0.672 Å. Although slightly higher than the inter-prediction comparisons, these values are still far below 1.5 Å, the commonly accepted threshold for near-identical protein structures. This indicates a very high level of accuracy, as the predicted conformations closely match the experimentally determined structure.

Interpretation:

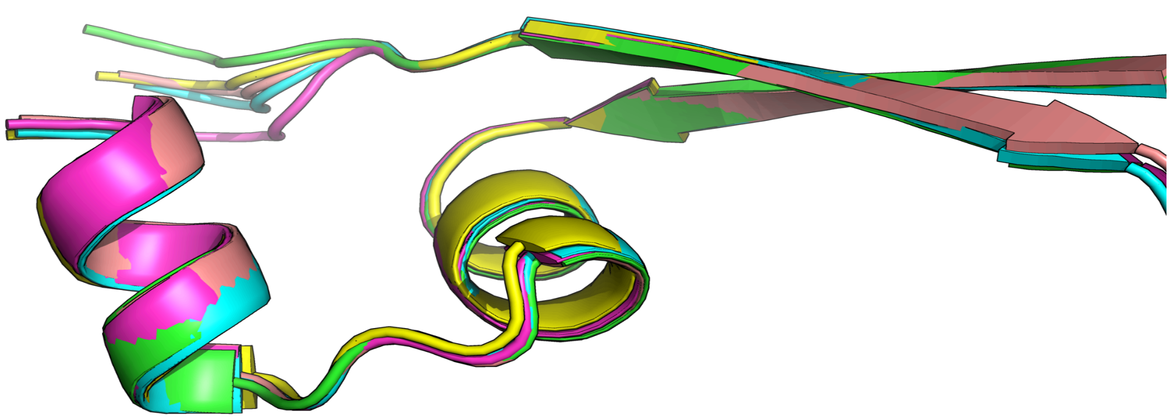
The combination of extremely high precision (strong agreement among predictions) and high accuracy (strong agreement with the Gold Standard) suggests that the prediction algorithm is both stable and reliable. The results demonstrate not only reproducibility across runs but also faithful structural representation of the experimentally determined protein structure. In summary, the RMSD matrix shows that the five predicted structures are tightly clustered and closely aligned with the experimental model. This highlights the strength and reliability of modern protein structure prediction methods such as AlphaFold in analyzing and modeling protein sequences.

6W6WB:

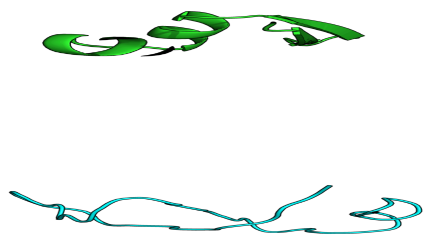
Information about the protein:
CST complex subunit CTC1|Homo sapiens (9606)
Sequence:AISQAIIRLLVEDGTAEAVVTCRNHHVAAALGLCPREWASLLD

Matrix obtained from alignment of all five sequences with each other and the Gold Standard
Note: All values are the Root-Mean-Square-Deviation(RMSD) measurement which uses Angstrom(Å) units

	S1	S2	S3	S4	S5	GS
S1	0	**.346**	**.317**	**.489**	**.342**	**12.307**
S2	.346	0	**.331**	**.414**	**.322**	**12.439**
S3	.317	.331	0	**.364**	**.284**	**12.629**
S4	.489	.414	.364	0	**.585**	**12.563**
S5	.342	.322	.284	.585	0	**12.352**



Caption: This image displays the superimposition of structures S1–S5 aligned on top of one another, clearly illustrating their extremely high structural similarity. The near-perfect overlap among the models visually confirms the strong precision observed in the RMSD analysis, demonstrating that the predicted structures are highly consistent with each other.



Caption:This image shows the best match (S1) aligned with the Gold Standard structure. Despite its high precision, the poor overlap highlights its low accuracy and large structural deviation.

Precision of the Predictions:

The RMSD values among S1–S5 are all very low, ranging from 0.284 to 0.585 Å. These values are well below 1 Å, indicating that all five predicted models are highly similar to one another. This demonstrates high precision, meaning the prediction method consistently produces nearly identical structures across multiple runs. For example, at the smallest deviation, S3 and S5 have an RMSD of 0.284 Å, showing very close agreement, while even the largest deviation among predictions (S4 vs S5 at 0.585 Å) remains small indicating that even those 2 have high similarity.Overall, the predictions cluster tightly together, suggesting strong reproducibility and internal consistency in the modeling process.

Accuracy Relative to the Gold Standard:

In contrast, the RMSD values between each predicted structure and the Gold Standard are very large (ranging from 12.3–12.6 Å). An RMSD of this magnitude indicates substantial structural deviation from the experimentally determined structure. This reflects low accuracy, meaning that although the predictions agree with each other, they do not closely match the true experimental conformation.Importantly, all five predicted structures show similarly high RMSD values relative to GS. This indicates that the deviation is systematic rather than random. In other words, the prediction method AlphaFold used consistently converges on a structure that differs significantly from the Gold Standard model.

Interpretation:

The combination of high precision and low accuracy suggests that the prediction algorithm is stable and reproducible but may have converged on an incorrect fold or orientation relative to the Gold Standard. In summary, the matrix demonstrates that the five predicted structures are highly precise (strong agreement among themselves) but not accurate (poor agreement with the Gold Standard). This distinction highlights the importance of evaluating both reproducibility and correctness when assessing protein structure prediction performance.

Conclusion:

In this project, two protein sequences were modeled using AlphaFold and compared to their experimentally determined structures from the Protein Data Bank. The first protein (PDB: 6Y4F chain A), demonstrated both exceptionally high precision and high accuracy. The five predicted models were nearly identical to one another (RMSD < 0.3 Å) and closely matched the experimental structure (RMSD ≈ 0.5–0.7 Å), indicating that AlphaFold successfully reproduced the correct three-dimensional fold with strong reliability.

In contrast, the second protein, (PDB: 6W6W chain B), showed high precision but low accuracy. Although the predicted models were tightly clustered (RMSD < 0.6 Å among predictions), all exhibited large deviations from the Gold Standard (RMSD ≈ 12 Å). This suggests that AlphaFold consistently converged on a stable but incorrect structural conformation for this sequence.

Overall, these results highlight an important distinction between precision and accuracy in protein structure prediction. AlphaFold demonstrated excellent reproducibility for both proteins, but accuracy depended on the specific sequence and structural context. The project illustrates both the strength of modern AI-based prediction methods and the continued importance of experimental validation when evaluating predicted protein structures.

Works Cited

Bank, Rcsb Protein Data. *1D PFV: 6Y4F*. www.rcsb.org/sequence/6Y4F#A.

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